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Citation

YUN, Jonghyuk; MOON, Jaeyoung; PARK, Yunseo; TAN, Sean Rui Xiang; KIM, Byunghyun; BALAN, Rajesh Krishna; and HAN, Jun. Hide-and-sweep: Detecting concealed cameras via LED illumination sweeps. (2026). *MobiSys '26: Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services, Cambridge, UK, June 21-25*. 750-765.
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Hide-and-Sweep: Detecting Concealed Cameras via LED Illumination Sweeps

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Abstract

Hidden cameras have increasingly infiltrated hotel and Airbnb rooms, posing serious privacy risks. Detecting such cameras is challenging because they are visually inconspicuous and often embedded inside everyday objects. Even worse, existing handheld detectors are manual and also rely on single-angle illumination and hence suffer from high false-positive rates. We present *SweepLED* (pronounced “sweepLED”)¹, a practical hidden camera detection system that operates on a commodity smartphone augmented with an unobtrusive LED-embedded case. *SweepLED* performs LED sweeping – a controlled sequence of multi-angle illumination – while the user simply holds the phone still by hand, enabling the camera to capture how reflections evolve under changing lighting. This reveals stable, lens-specific cues that distinguish hidden camera lenses from ordinary reflective objects, enabling robust detection with low user effort. We implement *SweepLED* using a compact hardware add-on and evaluate it in realistic environments containing 12 hidden-camera objects and 18 commonly reflective non-camera items. Our results demonstrate that *SweepLED* provides accurate and reliable hidden-camera detection using only unobtrusive smartphone-compatible hardware, achieving approximately 94% detection accuracy with a sweep time of under 5 s and a core component cost of less than USD \$7.

CCS Concepts

• Security and privacy; • Human-centered computing → Smartphones;

Keywords

Hidden Camera Detection; Privacy; Smartphone;

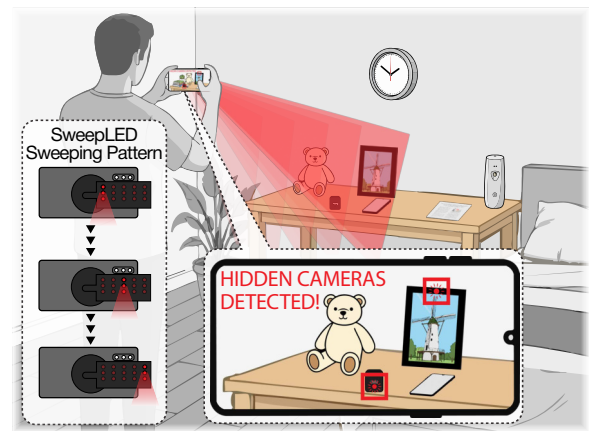


Figure 1: Overview of *SweepLED*. A smartphone with a low-cost LED-embedded case executes controlled illumination sweeps. By capturing reflections under changing angles while the user holds the phone still, *SweepLED* extracts lens-specific optical cues to separate hidden cameras from ordinary reflective objects.

ACM Reference Format:

Jonghyuk Yun, Jaeyoung Moon, Yunseo Park, Sean Rui Xiang Tan, Byunghyun Kim, Rajesh Krishna Balan, and Jun Han. 2026. Hide-and-Sweep: Detecting Concealed Cameras via LED Illumination Sweeps. In *The 24th Annual International Conference on Mobile Systems, Applications and Services (MobiSys '26)*, June 21–25, 2026, Cambridge, United Kingdom. ACM, New York, NY, USA, 16 pages. <https://doi.org/10.1145/3745756.3809236>

1 Introduction

Advances in miniaturized CMOS sensors and compact lenses have driven the rise of hidden cameras in lodging environments. Hotel rooms and Airbnb-style rentals are increasingly compromised by covert devices placed in everyday objects [2, 6, 11, 19, 23, 30, 50, 80]. With reports of guests being secretly recorded rising, reliable non-intrusive detection methods are urgently needed [20, 36, 54, 60].

A common consumer approach involves handheld detectors with built-in LEDs that require users to visually spot retro-reflections through a small viewfinder [4, 5, 46, 70]. In practice, these tools

¹Demo of *SweepLED* in action: <https://www.cyphy.kaist.ac.kr/research/sweepLED>



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ACM ISBN 979-8-4007-2027-7/26/06
<https://doi.org/10.1145/3745756.3809236>

are primarily used by law enforcement officers during targeted inspections, as they require slow manual scans coupled with careful visual scrutiny from a trained operator – because they use a **single illumination angle** that results in numerous everyday reflective surfaces, such as metal trims, glossy plastics, and glass, producing bright retro-reflections that appear indistinguishable from camera lenses. This causes *frequent false positives* and makes accurate identification extremely difficult, especially for ordinary users in hotel and rental environments.

To improve upon the limited accuracy of handheld detectors, researchers have proposed optical approaches that capture reflections under **multiple illumination angles**. The intuition is that by varying how light strikes a surface, lens-specific reflection behaviors may emerge that are not visible under single-angle illumination. A prominent example is LAPD [62], which leverages smartphone ToF hardware and its integrated laser emitter to scan the scene for retro-reflective responses from camera lenses. In practice, however, approaches like LAPD – and other systems that rely on sweeping a flashlight or LED source – perform multi-angle scanning by moving the *camera and light source together* as a single unit. We refer to this design pattern as **Camera–Light Coupled Multi-Angle Illumination**. However, this coupling introduces a limitation: because the camera and light source move simultaneously, any observed change in reflection may result either from altered illumination angles or from the device’s motion. These effects are inseparable when the system moves as one unit, producing ambiguous reflection patterns that hinder reliable discrimination between hidden camera lenses and ordinary reflective surfaces. Consequently, even multi-angle optical approaches struggle to deliver reliable detection accuracy in real lodging environments. This motivates the need for a fundamentally different approach.

To overcome the limitations of prior optical approaches, we pose a pivotal question: “**Can we achieve accurate hidden camera detection on an everyday smartphone by varying illumination without moving the camera?**” In response, we introduce *SweepLED* (pronounced “sweepLED”), a universally deployable approach that augments an ordinary smartphone with an unobtrusive LED-embedded phone case. As Figure 1 illustrates, the case performs a controlled LED sweeping sequence – a sequential LED illumination pattern – while the phone’s built-in RGB camera records the resulting reflections. During use, the user simply holds the phone at a few fixed viewpoints facing a suspicious object, pausing briefly at each position; while the user holds the phone steady at each viewpoint, the LED array continuously sweeps, producing multiple illumination angles. This process captures rich reflection behavior with low user effort and forms the input to *SweepLED*’s computer vision and learning pipeline. Note that our solution is robust against the motion jitter that arises from regular human hand activity (even when holding something still). The total time needed for capturing three different viewpoints is just 5 seconds or less. Our current prototype also uses an unobtrusive, cheap (less than USD \$7 for the core components) LED-embedded phone case that could be readily provided in hotel rooms or Airbnb rentals to assist guests in checking for hidden cameras.

Designing the LED-sweeping mechanism is integral to *SweepLED* because it enables **Camera–Light Decoupled Multi-Angle Illumination**, which is our key technique for reliably distinguishing

hidden camera lenses from ordinary reflective objects. By varying illumination angles while keeping the camera stationary, *SweepLED* exposes characteristic reflection behaviors: hidden camera lenses produce consistent, structured reflections due to their optical construction, whereas everyday reflective materials yield irregular or unstable responses (see §2.2). To exploit this effect, *SweepLED* performs LED sweeping while the smartphone is handheld, ensuring that reflection changes arise solely from illumination variation. These clean, lens-specific cues form the foundation of the system’s robust detection capability.

Designing *SweepLED* also introduces practical challenges beyond the core optical mechanism. Hidden cameras vary widely in shape, internal structure, and surface materials – ranging from glass lenses to acrylic covers to coated pinhole apertures – and are often embedded into heterogeneous surroundings such as glossy finishes, textured objects, or small openings. This diversity significantly alters how reflections appear, creating a broad range of optical patterns that the system must recognize robustly. To handle such variation, we design *SweepLED* to learn stable reflection patterns that persist across different lens types and environmental contexts, rather than relying on any single, fixed optical signature.

We implement and evaluate *SweepLED* through comprehensive real-world experiments, testing under varying conditions. Our evaluation uses 30 objects with and without hidden cameras, reflecting those commonly found in hotels and rentals. *SweepLED* achieves an overall detection accuracy of 93.9% with a per-object sweep time of under 5 seconds and a core component cost of less than USD \$7, demonstrating that *SweepLED* is a practical solution that performs reliably across diverse real-world settings. Overall, we make the following contributions:

- We propose *SweepLED*, a practical smartphone-based system for hidden camera detection using an unobtrusive LED-embedded case.
- We introduce and validate a **Camera–Light Decoupled Multi-Angle Illumination** strategy, supported by a vision and learning pipeline that extracts stable, lens-specific reflection cues.
- We implement a lightweight LED-embedded smartphone case that performs controlled illumination sweeps while maintaining a practical form factor.
- We demonstrate the effectiveness of *SweepLED* through comprehensive experiments with concealed camera objects and reflective everyday items in realistic environments.

2 Background, Core Idea, and Feasibility Study

We present relevant background information and the core idea of *SweepLED* followed by a feasibility study.

2.1 Optical Principles for Lens Detection

When light strikes a surface, it is reflected based on the surface’s properties and the geometry between three key players: (1) the *light source*, (2) the *reflector*, and (3) the *observer*. In our setting, the light source is an LED or flashlight, the reflector is any inspected object (e.g., a teddy bear’s glossy nose or a hidden-camera lens), and the observer is the human eye or the smartphone’s RGB camera.

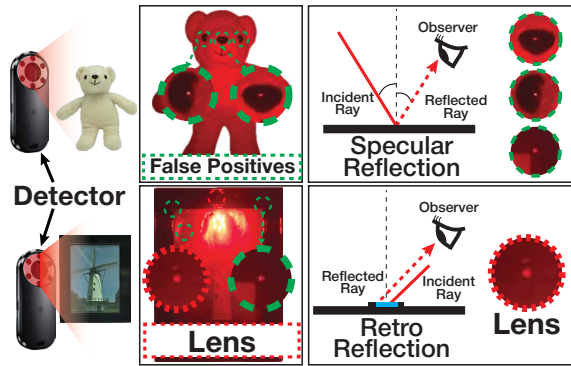


Figure 2: Comparison of specular and retro-reflection. Glossy objects produce specular highlights while camera lenses produce retro-reflective spots when illuminated by the handheld detector. These can appear visually similar, leading to false positives.

Specular reflection occurs on smooth, shiny surfaces. These reflections behave like a mirror: *the angle of reflection equals the angle of incidence*. Specular highlights often appear bright when the surface is oriented such that the reflected ray happens to reach the observer. Importantly, specular reflections are common on everyday objects such as plastic, metal trims, glass, and glossy coatings [72, 81].

Retro-reflection, in contrast, refers to *light reflected back toward the source*. True retro-reflection arises from specialized optical structures, such as *camera lenses*, whose curved elements and internal reflections redirect light toward the illumination direction over a wide range of angles [40–42]. These reflections appear as small, bright, stable highlights that remain visible even when the illumination angle changes.

However, **specular reflections can appear deceptively similar to retro-reflections** in a single snapshot. Under certain geometric alignments, ordinary surfaces may produce small, bright reflections that resemble the retro-reflective spots from camera lenses. Figure 2 illustrates this ambiguity: in a handheld detector view, shiny surfaces (e.g., a teddy bear’s eyes, a picture frame’s edges) can generate bright spots indistinguishable from those produced by actual lenses [71].

This ambiguity highlights a fundamental challenge: a single lighting angle is insufficient to reliably distinguish camera lenses from ordinary reflectors. The key lies in how **reflections change when the geometry between the light source, reflector, and observer is varied**. Camera lenses exhibit distinctive, stable behaviors under changing illumination, whereas everyday specular surfaces typically shift, smear, or vanish. This principle motivates multi-angle optical approaches, and ultimately motivates *SweepLED*’s camera–light decoupled design.

2.2 Core Idea

The central insight behind *SweepLED* is that camera lenses and ordinary glossy surfaces **respond differently when illumination changes while the camera remains fixed**. By isolating illumination variation from observer motion, *SweepLED* reveals reflection

patterns distinctive to hidden camera lenses. Figure 3 depicts this contrast. For a smooth, shiny surface such as a curved plastic teddy bear’s nose, the reflection follows simple specular behavior. As the illumination angle changes, the bright highlight simply *slides* across the surface like a mirror reflection. Its shape remains nearly unchanged – only its position shifts. Many everyday objects exhibit this kind of moving but structurally stable highlight.

Hidden camera lenses behave fundamentally differently. A lens–aperture–sensor assembly not only produces retro-reflection but also introduces *geometric vignetting*: when the illumination tilts, parts of the incoming or outgoing light paths become blocked by the aperture or barrel. As a result, the reflection does not simply move – it *changes shape*, often forming “cat-eye” or “crescent-like” patterns [58]. These structured distortions are intrinsic to camera optics and do not arise from ordinary reflective materials.

Prior optical methods that move the camera and light source together mix illumination changes with viewpoint changes, obscuring the differences between reflectors. By keeping the camera fixed and sweeping only the illumination, *SweepLED* cleanly separates these effects, making the contrast between ordinary specular reflections and lens-specific distortions explicit. This controlled separation is the core optical basis of *SweepLED*’s detection capability.

2.3 Feasibility Study

We conduct a feasibility study to validate our core idea by comparing a single-angle illumination setting with our multi-angle illumination sequence using the same prototype hardware. Using this prototype hardware (see §4), we collect data from 12 distinct objects containing hidden camera lenses, along with a diverse set of ordinary reflective surfaces. For the single-angle illumination, we consider the brightest point under a fixed illumination angle. For the multi-angle setting, we sweep the illumination angle and analyze the resulting sequence of reflection responses. Figure 4 visualizes the resulting feature distributions generated using the t-SNE [32, 47] statistical method to compare embeddings extracted under single-angle illumination with those obtained from our multi-angle illumination sequence (i.e., *SweepLED*’s approach). For this analysis, we extract features from each sample using a pre-trained ResNet-18 [15, 27].

Figure 4(a) shows that the embedded features exhibit substantial overlap between objects with and without hidden cameras, highlighting the ambiguity inherent in single-angle illumination. In contrast, as Figure 4(b) shows, incorporating angular diversity produces much tighter clustering amongst hidden cameras and clearer separation from other reflective surfaces. These results confirm that the temporal signatures induced by changing illumination angles provide critical discriminative information for reliable lens detection.

However, relying solely on generic deep features also exposes a key limitation in generalization. For example, *Object 9* forms a tight intra-object cluster but still lies close to the false-positive group in the embedding space as the variance across different hidden cameras can be high. As such, *SweepLED* uses a specialised framework, explained in the next section, to reduce this variance by explicitly capturing reflection changes caused by varying illumination angles.

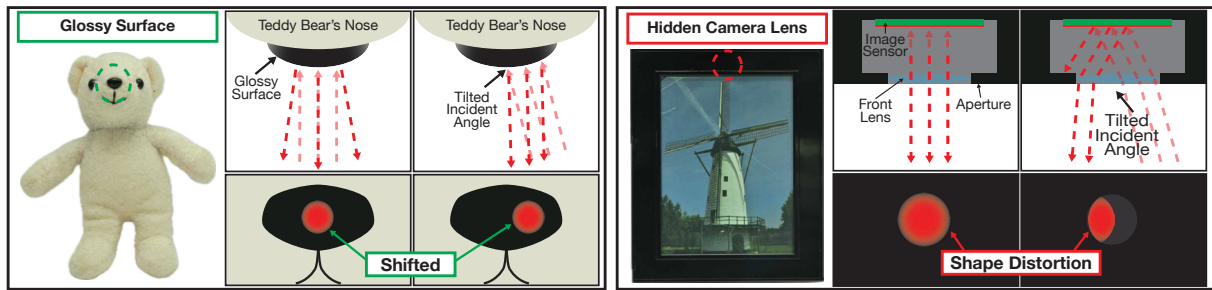


Figure 3: Reflection behavior under changing illumination angles. The left side highlights on smooth, shiny surfaces that shift position as the light direction changes, while retaining a similar appearance. The right side shows reflections from a camera lens that deform noticeably under varying illumination, revealing shape changes.

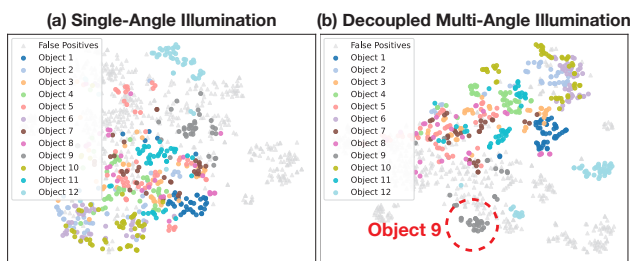


Figure 4: Figure depicts *SweepLED*'s feasibility study. (a) yields overlapping embeddings between lenses and non-lens reflections. (b) yields tighter clustering and clearer separation, enhancing discriminability.

2.4 From Feasibility to Solution

From our prior work search and feasibility study, several key requirements emerge: (a) *high accuracy*, (b) *ease of use*, (c) *reasonably fast*, and (d) *low cost*. Achieving high accuracy, in particular, necessitates a camera–light decoupled strategy with multi-angle illumination.

After exploring several alternatives, we converge on a smartphone case–style attachment containing an LED array, which provides the multi-angle illumination required for accurate detection. As shown in Figure 5(a), the array attaches to the phone like a standard case. During use, the user briefly positions the phone at three viewpoints around a suspected object while the LEDs sweep the scene. The phone camera records the resulting reflections, and our software analyzes the video to determine whether a hidden camera is present.

This design meets the accuracy requirement (see §5) while remaining practical: the LED array attaches like a normal phone case, each scan completes in under 5 seconds, and the core hardware components cost less than \$7.

3 System Design

We now present the system design of *SweepLED*.

3.1 System Overview

SweepLED identifies hidden cameras by exploiting the distinctive temporal reflection patterns that camera lenses produce under controlled, angle-varying illumination. A compact LED-array phone case generates these patterns, and *SweepLED*'s deep-learning modules are trained on labeled footage of diverse hidden camera types. During deployment, the user simply pans a *SweepLED*-enabled smartphone across a scene while the system probes with the LED array and analyzes the captured video to flag potential lenses.

Figure 5 presents an overview of *SweepLED*'s processing pipeline. During each LED sweeping sequence, the system records video while sequentially activating LEDs to sweep illumination across the scene. In *Preprocessing* (§3.2), *SweepLED* converts the raw video into contrast-enhanced difference frames aligned with the illumination pattern. *Candidate Region Detection* (§3.3) then localizes “lens-like” reflections using a high-recall detector. These regions are refined by *Cross-Frame Detection Refinement* (§3.4), which tracks reflections over time and removes transient artifacts. Finally, *Lens Classification* (§3.5) analyzes each region's spatial and temporal response to determine whether it corresponds to a hidden camera lens.

3.2 Preprocessing

Preprocessing reduces environmental noise to enable robust hidden camera reflection detection. Although true reflections appear as bright regions, ambient lighting, surface texture, and scene geometry often introduce many spurious highlights, and handheld operation introduces additional motion noise. To address this, preprocessing stabilizes the raw video, isolates illumination changes driven by *SweepLED*'s LED sweep, and outputs noise-reduced *difference frames* suitable for reliable downstream detection.

Video Stabilization. To compensate for motion artifacts inherent to handheld operation (due to jitters and hand tremors), we estimate inter-frame transformations using the Lucas–Kanade method [45, 67] and warp each frame to a reference coordinate system. This alignment removes unintended motion so that the remaining temporal differences more reliably reflect true illumination changes.

Difference Frame Extraction. We pair stabilized frames with the known illumination waveform to identify baseline (LED-off) and illuminated (LED-on) states. As the scene remains relatively static between probes, computing the *difference* between these frames

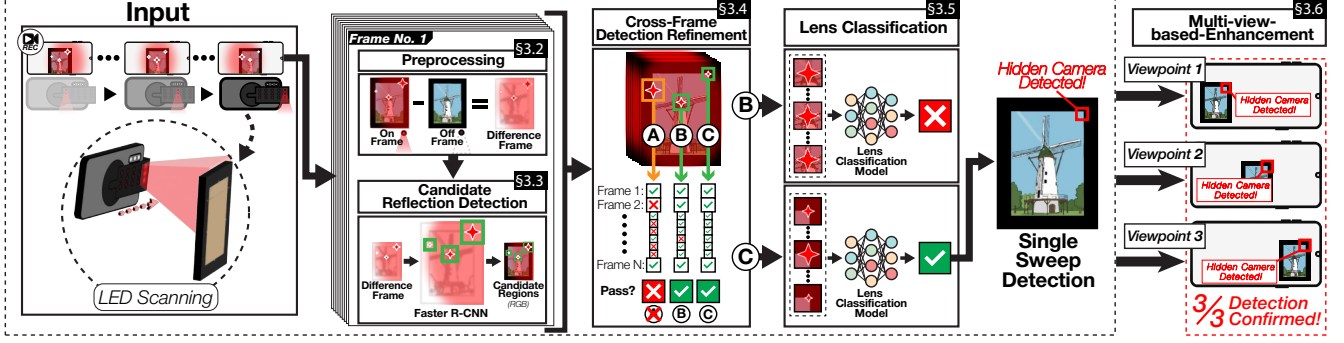


Figure 5: Overview of *SweepLED*'s processing pipeline, which takes a short video of a scene illuminated by LED sweeping sequence as input. *Preprocessing* (§3.2) isolates sweep reflections in noise-reduced *difference frames*. *Candidate Region Detection* (§3.3) localizes potential targets, which are filtered by *Cross-Frame Detection Refinement* (§3.4) to retain only *persistently reflective regions*. *Lens Classification* (§3.5) performs fine-grained reflection analysis to identify hidden cameras. *Multi-view-based Enhancement* (§3.6) fuses a few viewpoints for enhancement.

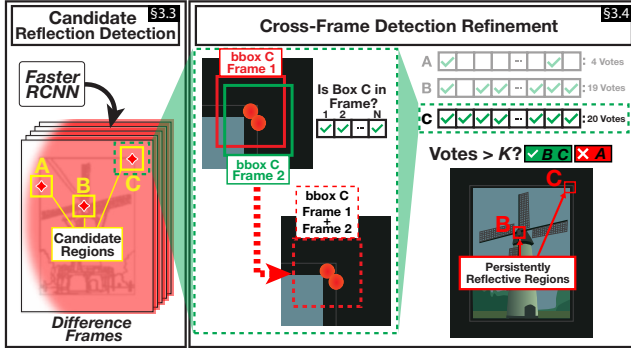


Figure 6: Candidate Reflection Detection (§3.3) uses *difference frames* to a detection model trained to find small, circular, “*lens-like*” reflections, outputting bounding boxes of candidate regions. *Cross-Frame Detection Refinement* (§3.4) tracks reflections across multiple frames by fusing over time if bounding boxes overlap ($\text{IoU} > \text{Threshold}$), then the frames in which a reflection appears. Only *persistently reflective region* (i.e., appear in K out of N frames) frames are retained.

(i.e., *difference frames*) highlights LED-induced reflections while suppressing noise.

Contrast Amplification. We apply Contrast Limited Adaptive Histogram Equalization (CLAHE) [55, 66, 86] on each *difference frame*. This improves local contrast and preserves structural details, making probing reflections easier to detect before forwarding each *difference frame* to the next module.

3.3 Candidate Reflection Detection

This module aims to detect regions in each input *difference frame* that correspond only to “*lens-like*” reflections, while ignoring spurious reflections from other objects. Since *SweepLED*'s LED sweeping sequence equally illuminates all objects in its field-of-view, both

hidden cameras and other reflective objects might produce strong reflections.

To account for this, we leverage the observation that hidden cameras produce reflections with bounded spatial properties (i.e., small size, circular or elliptical shape), while reflections from most other surfaces exhibit unconstrained and random spatial distributions. Hence, during bootstrapping, we employ a Faster R-CNN object detection model to learn these properties from *difference frames* annotated with hidden camera positions [56, 57]. During deployment, the detection model outputs bounding boxes and confidence scores of candidate regions with “*lens-like*” reflections. Figure 6 depicts the resulting candidate regions, annotated as A, B, and C. To mitigate the effect of unseen camera configurations leading to missed detections, we calibrate this initial detection step to reduce missed detections by adopting a loose detection threshold and relying on subsequent modules to refine these initial detection results.

3.4 Cross-Frame Detection Refinement

This module takes as input the candidate reflection regions (e.g., A, B, and C depicted in Figure 6), tracks each reflection over time (i.e., across frames), and outputs a list of *persistently reflective regions*. While hidden cameras exhibit strong reflections throughout the LED sweeping sequence, other objects may also momentarily appear “*lens-like*” at certain points in the sequence (i.e., at illumination-specific angles), leading to false detections.

To account for this, we first combine all candidate regions across the N frames and project them into a common coordinate system corresponding to a single reference *difference frame*. Next, using Non-Maximum Suppression, we fuse regions across frames with significant overlap (i.e., their intersection-over-union (IoU) exceeds a predefined threshold), keeping only regions that persist in K out of N frames. We depict this process in Figure 6 for reflection C. Each region (i.e., bbox C/Frame 1 and bbox C/Frame 2) is projected into the same coordinate space, then fused into a single bounding box. Finally, we count the number of frames in which reflection C was detected, and only keep it if it appears in at least K frames.

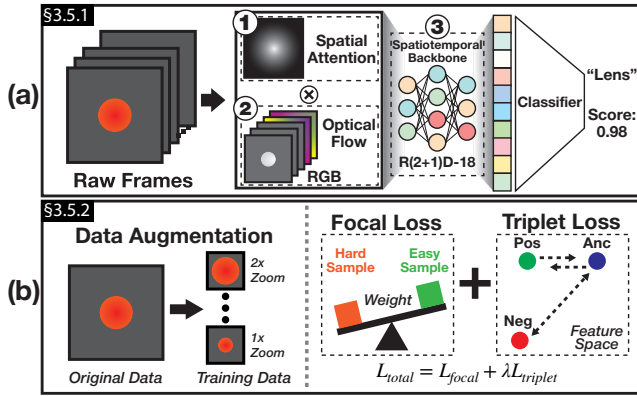


Figure 7: (a) depicts Classifier Architecture Design (§3.5.1), with Spatial Attention, RGB Optical Flow fusion, and Spatiotemporal Backbone to extract fine-grained reflective patterns for lens vs. non-lens classification. (b) depicts Training Strategy (§3.5.2), with Data Augmentation, Focal and Triple Losses jointly improving the robustness.

Reflections **A** and **B** also undergo the same process, but **A** is discarded for not meeting the persistence criteria, while **B** and **C** are retained.

Each fusion results in a region-of-interest that encapsulates (i.e., tracks) a persistent “lens-like” reflection (i.e., *persistently reflective region*) over the period of *SweepLED*’s LED sweeping sequence, and takes on the maximum confidence score of its constituent sub-regions. The list of *persistently reflective regions* are passed to the next module for fine-grained analysis.

3.5 Lens Classification

The goal of this module is to effectively differentiate between hidden camera lens and non-lens reflections. This module takes in the list of *persistently reflective regions*, and classifies each into hidden camera and non-lens (i.e., false detection) regions, producing as output the final list of suspected hidden cameras. Each *persistently reflective region* consists of either: (1) an actual hidden camera; or (2) an object with strong “lens-like” reflective properties.

We observe that such non-lens reflective objects frequently have curved, “lens-like” geometries (e.g., LEDs, raise shiny “bumps”) or have reflective surfaces recessed in small holes or gaps. The resulting reflections are visually similar to reflections from hidden cameras and are challenging to distinguish from a single snapshot. Hence, during the LED sweeping sequence, the changes in illumination angle result in subtle, fine-grained changes to the size, shape, intensity, and location of each reflection over time.

To effectively differentiate lens and non-lens reflections from these fine-grained changes, we (1) design our lens classifier’s architecture for holistic capture of both camera-related local (i.e., between consecutive frames) and global (i.e., across multiple frames) patterns in the changes, and (2) train the classifier to generalize across diverse scenarios even when the observable difference in patterns is small. Figure 7 presents the architecture and training strategy in detail, which we discuss below.

3.5.1 Classifier Architecture Design. The Lens Classification module takes as input the RGB video stream for each *persistently reflective region*, extracts features that encode reflection changes, then classifies them to produce a binary hidden camera or non-camera result. Figure 7(a) presents the three major components used to extract features specific to camera/non-camera classification.

① **Spatial Attention.** Since each *persistently reflective region* is centered around a single potential hidden camera, valid retro-reflective signals must appear near the center of the frame. Hence, we impose a static spatial attention mechanism with a 2D Gaussian mask M ($\sigma = 0.5$) centered at the region of interest [37]. This mask is applied as element-wise multiplication to the input volume, $X_{masked} = X \odot M$. By attenuating surrounding pixels where lens reflections are optically implausible, the model suppresses irrelevant background motion and improves classifier stability across varied environments.

② **Optical Flow Fusion Input.** Without guidance, the model struggles to capture subtle temporal gradients (i.e., small changes or movements) in reflections from the RGB video stream alone. To handle this, we make temporal motion cues explicit by adding dense optical flow fields between consecutive frames to the input. Specifically, we calculate the optical flow of grayscale frames with Farneback methods [22, 73], and concatenate it with RGB frames. These motion cues aid the model in extracting subtle short-term dynamics in the scene (i.e., small changes between each frame).

③ **Spatiotemporal Backbone.** Although the short-term dynamics are captured, the global reflective response of lens-and-sensor assembly as incident angle varies over time (i.e., overall changes in reflection) also carries important information that can help distinguish hidden camera from spurious reflections. Hence, we process the volume with a R(2+1)D-18 backbone [9, 35, 75, 76], which decomposes 3D convolutions into separate spatial and temporal operations. The backbone complements this limitation by modeling longer-range temporal dynamics and aggregating multi-frame intensity patterns to fully capture both local and global aspects of the reflective signal the entire LED sweeping sequence.

3.5.2 Training Strategy. Our classifier needs to distinguish retro-reflective lens signatures from diverse background objects. This task is challenging because real-world environments exhibit substantial variability in surface materials, lighting conditions, and object geometries. Furthermore, even with our model architecture, the differences between reflections for hidden camera and non-cameras can be minute and difficult to learn.

Figure 7(b) shows the key training techniques that we employ. First, we augment training data to simulate lenses of random sizes via cropping the central region of the image. This augmentation exposes the model to a broad spectrum of scale variations that naturally arise due to different capture distances [68]. To promote robust generalization during training, we also employ a training strategy that integrates two well-known techniques: focal loss [12, 39, 49] and triplet-based metric learning objective [13, 26, 31, 65]. Training with Focal loss allows the model initially identify hard-to-classify examples (e.g., those with subtle lens reflections) and prioritize them by dynamically down-weighting the loss contribution from easy-to-classify background samples in subsequent training cycles. Meanwhile, Triplet-based metric learning objective allows the

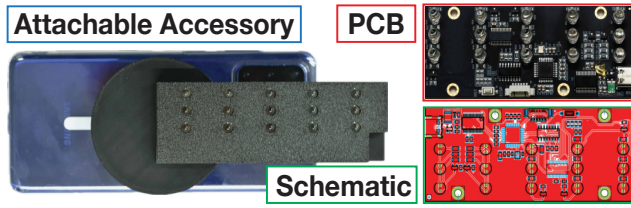


Figure 8: *SweepLED*'s prototype hardware: an unobtrusive attachable phone-case accessory with directional LED array. The compact module offers stable multi-angle illumination, supporting under 5s per-object scans with just \$7.

model to learn a discriminative embedding space where similar samples (e.g., all hidden camera reflections) are mapped close to each other, while dissimilar samples (e.g., hidden camera vs. non-camera) are pushed far apart. Together, these techniques allow the model to prioritize informative samples, learn scale-invariant representations, and form a discriminative embedding space that effectively separates true lens reflections from visually similar artifacts.

The final output of each classification is a binary classification for each input *persistently reflective region*, indicating whether it is a hidden camera or non-camera (false detection). This produces the final list of suspected hidden cameras for a single LED sweeping sequence, which is returned to the user.

3.6 Multi-view-based Enhancement

In practical scenarios, detecting hidden cameras in the wild with *SweepLED* presents additional difficulties. First, strong and direct (i.e., head-on) illumination from *SweepLED* potentially saturates the region directly in front of the smartphone with high intensity reflections, obscuring hidden cameras in those regions from detection. Next, the compactness of the hardware attachment limits the range of illumination angles during LED sweeping sequence; some reflections appear identical to hidden camera within the limited set of illumination angles, which leads to false detections.

To account for these cases, we design *SweepLED* to enable users to combine results for multiple LED sweeping sequence from different viewpoints, as depicted at the last step in Figure 5. Between each sweep, users shift the smartphone by a short distance (e.g., 3 cm). *SweepLED* finalizes the decision using a majority voting mechanism; if each sweep consistently detects a hidden camera the same physical location in a majority of the views (e.g., 2 out of 3 scans), *SweepLED* reports the object as a hidden camera.

4 Prototype Implementation

As depicted in Figure 8, we prototype our hardware as a compact and attachable accessory. The device incorporates five sequences of LED to generate multiple illumination angles. Since the narrower beam angles increase the signal-to-noise ratio of the reflected signal and enable clearer separation of optical illumination directions, we employ VLCS5830 LEDs [79], which provide a beam profile of approximately $\pm 4^\circ$. To produce smooth illumination transitions that appear as continuous motion, we adopt a PWM-based duty-cycle control scheme powered by the PCA9685 PWM driver [52]. All

operations, such as LED sequencing, PWM modulation, and timing control, are controlled by an ATmega328P microcontroller [8]. We also implement a custom Android application to control the microcontroller and record camera videos. Specifically, we control the microcontroller by toggling the Data Terminal Ready (DTR) line, which triggers the execution of the onboard PWM control sequence. During illumination, we record videos using CameraX with Camera2 interoperability, fixing the frame rate, ISO, and shutter speed to ensure consistent exposure parameters. This compact, modular design enhances user convenience by attaching like a standard phone case while providing stable, directionally controlled illumination; it also supports under 5s per-object scanning and uses core components costing under USD \$7, making it practical for real-world deployment.

5 Evaluation

We now present the real-world performance of *SweepLED*.

5.1 Experimental Setup

Apparatus. Figure 9(a) shows our evaluation setup. We collect videos from 12 commercially available hidden-camera objects purchased online (e.g., Amazon, AliExpress), covering a diverse range of real-world form factors. To broaden the scope, we also include 18 non-camera objects frequently mentioned in crime reports, focusing on items with holes or glossy surfaces to more rigorously test robustness.

Data Collection. We conduct all experiments in real indoor environments under consistent lighting. Data collection and model training use two smartphones: a Samsung Galaxy S20+ and a Galaxy Z Fold7 [64], recording UHD video at 30 fps [17, 18]. To ensure consistent exposure, we fix the shutter speed (1/60) and ISO (150) on both devices [16].

For training the Candidate Reflection Detection (§3.3) and Lens Classification (§3.5) modules, we collect over 7,300 annotated images and 63,000 cropped videos (3,000 lens, 60,000 non-lens). To evaluate generalizability, we adopt a Leave- N -Out strategy, excluding each test object from training. All evaluation experiments use a single phone (S20+), except where cross-device comparisons are performed. We conduct all experiments in compliance with approval from our Institutional Review Board (IRB).

Evaluation Metric. We consider a detection successful when the predicted top-1 region matches the ground-truth location in more than two of three consecutive frames. Under this criterion, each trial is labeled as follows: a *True Positive* when *SweepLED* correctly identifies a lens in an object with a hidden camera; a *True Negative* when it reports no detection for an object without a camera; a *False Negative* when it misses a lens in a camera-containing object; and a *False Positive* when it reports a lens for an object without a camera or at the wrong location.

5.2 Overall Performance

We evaluate the overall performance of *SweepLED* using handheld smartphone captures to reflect realistic user behavior, without a tripod or stand. All results are reported on unseen objects using a Leave- N -Out training strategy.

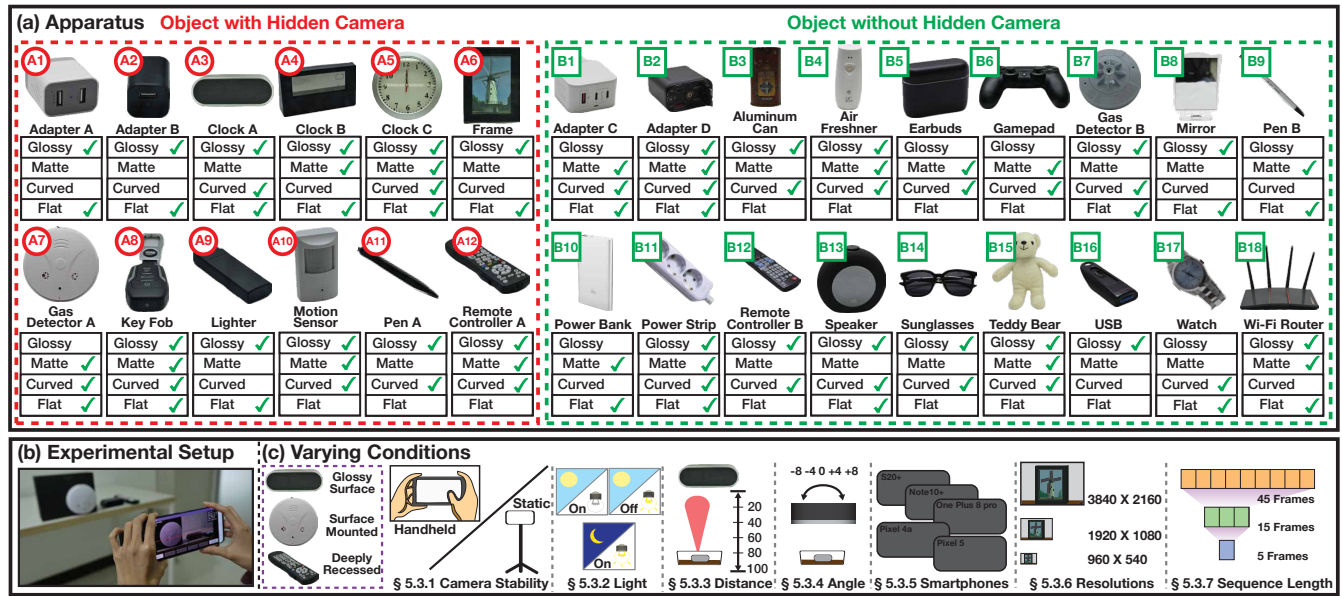


Figure 9: Figure depicts the setup of our experiments. (a) The 30 objects used in our evaluation, consisting of 12 objects containing hidden cameras and 18 without. We characterize each object along two surface dimensions, reflectivity and geometry. We assign the labels based on the hidden camera region and nearby optical decoys; for objects without hidden cameras, we consider the dominant surface characteristics and the most camera-like decoy region. (b) Our experimental setup using *SweepLED*'s prototype hardware to scan multiple objects. (c) Our comprehensive evaluation under varying experimental conditions.

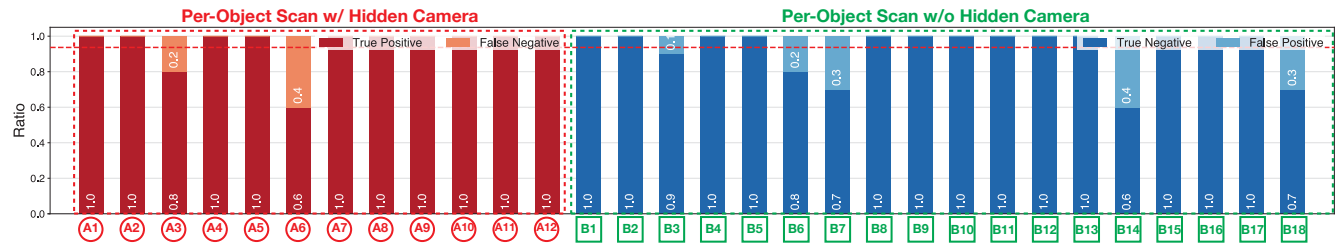


Figure 10: Figure depicts the *per-object* scanning overall results, including the *TP* and *FN* results for objects with hidden camera; and *TN* and *FP* rates for objects without camera. The dotted line indicates the average accuracy across objects (93.9%).

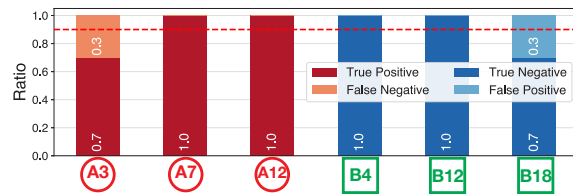


Figure 11: Figure depicts the *multi-object* scanning overall results, including the *TP* and *FN* results for objects with hidden camera; and *TN* and *FP* rates for objects without camera. The dotted line indicates the average accuracy across objects (90.0%).

5.2.1 Per-object Overall Results. We first evaluate per-object performance. Following the multi-view enhancement design (§3.6),

SweepLED performs LED sweeping from three viewpoints: one central view and two rightward lateral shifts. Each sweep lasts 1.5 seconds, totaling 4.5 seconds per object. We apply this procedure to all 30 objects (10 trials each), as shown in Figure 9(a). To further understand factors affecting the performance, we characterize the 30 objects along two surface dimensions: *reflectivity* and *geometry*. We assign these attributes by considering the surfaces at the exact camera location as well as nearby regions that may induce confusion, such as occluding structures or optical decoys. For objects containing hidden cameras, such decoys include reflective components or geometrically camera-like parts within the object. For objects without hidden cameras, we consider both the dominant surface characteristics and the properties of the most camera-like decoy regions. Using this characterization, we examine how surface reflectivity and geometry influence *SweepLED*'s performance.

Figure 10 shows that *SweepLED* achieves an overall accuracy of 93.9%. On the 12 hidden-camera objects, it detects an average of 95%, while producing only a 7.2% false-positive rate on objects without cameras. The main failure case is object **A6**, which attains 60% accuracy due to strong specular reflections saturating the sensor and overpowering the weaker retro-reflection from the lens. In practice, slightly tilting the phone or adjusting the viewpoint vertically reduces surface glare would further help the detection.

We also observe higher false-positive rates for **B7**, **B14**, and **B18** (30%, 40%, and 30%). These cases arise from surface features that produce highlights resembling lens retro-reflections. For example, **B7** and **B18** contain reflective metallic regions recessed in cavities, and **B7** consistently generates a strong corner highlight that is misclassified despite its differing shape. Importantly, the overall false-positive rate remains low, and in safety-critical settings like hidden-camera detection, avoiding missed detections is typically more important than eliminating all false alarms. A small number of false positives is therefore acceptable, enabling the system to remain conservative and sensitive to potential threats. To further assess performance beyond the 30 objects, we additionally evaluate *SweepLED* on a more diverse set of objects in Appendix A.

5.2.2 Multi-object Overall Results. We evaluate the capability of *SweepLED* to scan multiple objects in a single pass, as shown in Figure 9(b). We use two sets of three objects – one with hidden cameras (**A3**, **A7**, **A12**) and one without (**B4**, **B12**, **B18**) – selected to cover both easy and challenging cases. Using two groups of three reflects realistic hotel-room usage, where users typically inspect small clusters of nearby items rather than a densely packed set of many objects. As noted in §3.6, *SweepLED* improves accuracy by sweeping from three viewpoints per object. When multiple objects share a scene, many viewpoints overlap, allowing us to reduce the total required views. In our multi-object setups, each three-object group is scanned using only six viewpoints, yielding much faster inspection. Each trial consists of six illuminated sequences (1.5 s each), totaling roughly 9 s per scan. We repeat this process for 10 independent trials per group.

Figure 11 summarizes the results: across the two three-object groups, *SweepLED* achieves 100% accuracy on four objects and 70% accuracy on the remaining two. These outcomes closely mirror the trends observed in the per-object evaluation, confirming that *SweepLED* can effectively scan multiple items at once while preserving overall accuracy and reducing the total time required for inspection.

5.3 Differing Conditions

Figure 9(c) shows the setup for our controlled evaluation. To assess *SweepLED*'s robustness, we test its performance under diverse environmental and experimental conditions using three representative hidden-camera objects: a glossy reflective surface (**A3**), a deeply recessed lens (**A12**), and a surface-mounted miniature lens (**A7**). In all experiments, each object is positioned so its lens appears in the upper region of the frame, and every condition is evaluated over 10 trials per object.

5.3.1 Varying Camera Stability. To evaluate the impact of natural hand motion – such as jitter or minor tremor – we compare

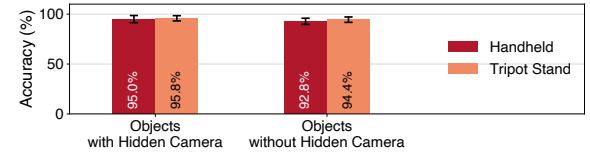


Figure 12: Figure depicts the comparison of handheld (*SweepLED*'s design) vs. smartphone on tripod stand.

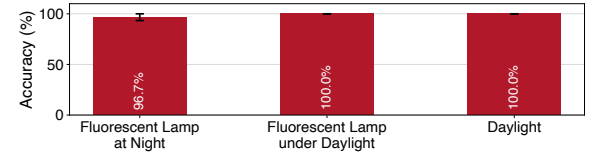


Figure 13: Figure depicts the accuracy across varying indoor light settings.

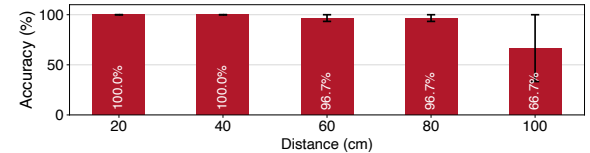


Figure 14: Figure depicts the accuracy across varying detection distances.

SweepLED's performance using handheld captures against captures taken with a tripod. As shown in Figure 12, accuracy remains comparable in both conditions (93.9% handheld vs. 95.1% tripod), demonstrating robustness to user-induced motion. This stability is aided by the *Video Stabilization* step in the *Preprocessing* module (§3.2).

5.3.2 Varying Lighting Conditions. Since *SweepLED* relies on optical sensing, its performance may vary with ambient lighting. We evaluate robustness under three representative indoor conditions, including nighttime with ceiling lights, daytime with ceiling lights, and daytime with natural light only. As shown in Figure 13, *SweepLED* maintains at least 96.7% accuracy across all settings, indicating that typical indoor lighting variations have limited impact and *SweepLED* remains reliable in everyday environments.

5.3.3 Varying Detection Distances. We evaluate *SweepLED* at different capture distances. As shown in Figure 14, accuracies of 100%, 100%, 96.7%, 96.7%, and 66.7% are achieved as distance increases. *SweepLED* remains reliable up to 80 cm, but performance drops at 100 cm. At this range, the Faster R-CNN detector fails to localize the hidden camera in **A12**, likely because subtle lens reflections become difficult to distinguish from other reflective surfaces at longer distances.

5.3.4 Varying Lens Orientation Angles. In real deployments, hidden cameras are often installed with lenses that are not perfectly front-facing – they may be slightly tilted due to intentional concealment, mounting constraints, or manufacturing tolerances. To evaluate *SweepLED*'s robustness to such installation variations, we assess

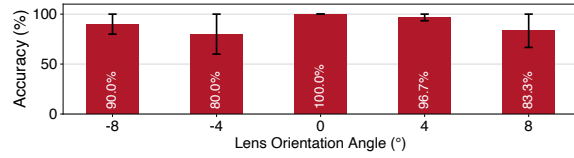


Figure 15: Figure depicts the accuracy across varying lens orientation angles.

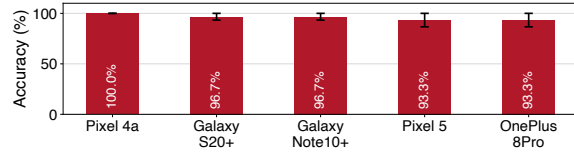


Figure 16: Figure depicts the accuracy across different smartphone models from diverse vendors.

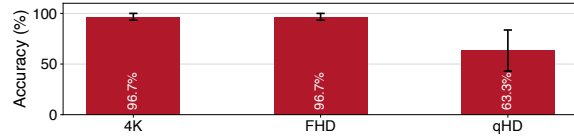


Figure 17: Figure depicts accuracy across different video resolutions.

performance across a controlled range of lens orientations. We test five angles from -8° to 8° in increments of 4° . Figure 15 depicts *SweepLED*'s detection performance under these conditions. We achieve peak accuracy of 96.7% when viewing objects at 0° , while performance notably drops at -4° due to a failure case involving object **A3**. At this angle, a strong direct surface reflection saturates the lens region, preventing the inference module from observing the temporal changes needed for classification. At larger tilt angles, the retro-reflection becomes dimmer and more distorted, reducing the available signal for robust detection.

5.3.5 Varying Smartphone Models. Given the diversity of smartphone camera models, variations in sensor quality and lens design can lead to differences in dynamic range and noise characteristics. Moreover, vendor-specific image signal processing can introduce additional discrepancies in the recorded optical responses across devices. We evaluate *SweepLED*'s generalizability across heterogeneous smartphones by testing five devices with distinct camera specifications. We use the Galaxy S20+ as our primary development device, while the remaining four devices are never seen during training. As depicted in Figure 16, *SweepLED* achieves 96.7%, 96.7%, 100%, 93.3%, and 93.3% accuracy on the Galaxy S20+, Galaxy Note 10+, Pixel 4a, Pixel 5, and OnePlus 8 Pro, respectively [29, 53, 63]. These results show that *SweepLED* maintains consistently high detection performance not only on the training device but also across unseen smartphone platforms with diverse hardware and imaging pipelines.

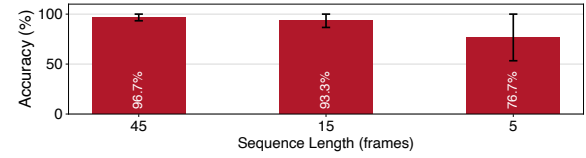


Figure 18: Figure depicts accuracy across different sequence lengths.

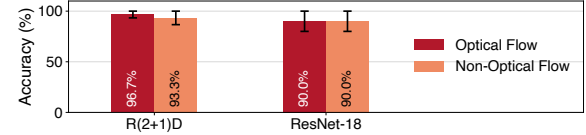


Figure 19: Figure depicts the comparison of different architectural choices.

5.3.6 Varying Video Resolution. Although modern smartphones are increasingly capable of capturing 4K (UHD) video, some low-end devices remain incompatible with this configuration. In addition, operating at lower resolutions can significantly reduce computational cost and improve responsiveness. To evaluate *SweepLED*'s robustness to different video resolutions, we take the same 4K recordings and generate three versions: the original 4K, a downsampled Full HD (1920×1080), and a downsampled qHD (960×540) resolution, where the lower-resolution videos are then upscaled back to 4K before being input into our pipeline. As depicted in Figure 17, *SweepLED* achieves detection accuracies of 96.7%, 93.3%, and 63.3% at 4K, Full HD, and qHD, respectively, indicating that *SweepLED* remains robust at 4K and Full HD but degrades noticeably when only very low-resolution video is available. These results suggest that severe loss of spatial information removes fine reflection details that *SweepLED* relies on, directly impacting detection performance. Since most commodity smartphones already support at least 1920×1080 video recording, our system is practically deployable on a wide range of existing devices.

5.3.7 Varying Sequence Length. To understand how *SweepLED* performs under faster or shorter LED scanning motions, we evaluate its robustness when the illumination sequence is reduced in length. In this experiment, we vary the sequence length by sub-sampling our original 45-frame sequence, while keeping the source frame rate and duration constant. We train models where the full LED sweeping sequence is captured within 45-, 15-, and 5-frames. As shown in Figure 18, *SweepLED* achieves detection accuracies of 96.7%, 93.3%, and 76.7% for 45-, 15-, and 5-frame sequences, respectively, showing stable performance above for sequences above 15-frames. This implies redundancy between frames, as shorter sequences can fully capture the changes in reflection that are critical to *SweepLED*, which indicates the potential for reducing the duration of the LED sweeping sequence.

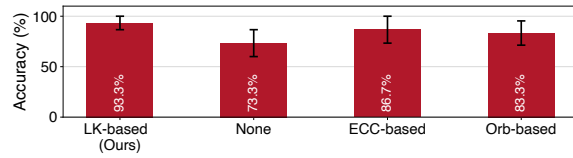


Figure 20: Figure depicts accuracy across different video stabilization methods.

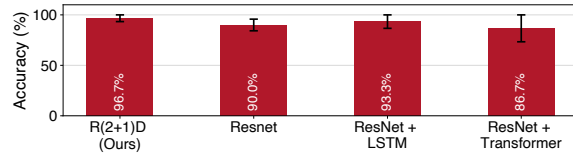


Figure 21: Figure depicts accuracy across different spatio-temporal classifier architectures.

5.4 Comparing Architectural Choices

We analyze how different components of our learning pipeline contribute to detection accuracy. We compare two backbone families – a 2D CNN (ResNet-18) and a spatio-temporal model (R(2+1)D) – under configurations with and without optical-flow features. As illustrated in Figure 19, the best-performing variant combines R(2+1)D with optical flow (96.7%). This combination captures both the fine-grained motion induced by LED sweeping and the longer-range temporal patterns that distinguish lens behavior from ordinary reflections. We note that ResNet-18 shows minimal sensitivity to optical flow, which is expected since it processes frames independently and cannot leverage temporal dynamics. Importantly, all configurations achieve reasonably strong performance, suggesting that the optical signature exploited by *SweepLED* does not rely heavily on any single architectural choice.

5.5 Module Evaluation

To further clarify the design choices of our system, we conduct additional module evaluations on two key components: the video stabilization module and the spatio-temporal classifier.

Video stabilization. We evaluate the impact of different video stabilization methods, including no stabilization, Enhanced Correlation Coefficient (ECC)-based stabilization [21], Oriented FAST and Rotated BRIEF (ORB)-based stabilization [59], and our Lucas-Kanade-based approach [45]. As shown in Figure 20, our method achieves a detection accuracy of 93.3%, outperforming no stabilization (73.3%), ECC-based stabilization (86.7%), and ORB-based stabilization (83.3%). We find that ECC-based and ORB-based stabilization often fail to produce stable alignment when the projected light pattern contains high-intensity reflections and rapid shadow changes. Such conditions make feature matching and image alignment less reliable, which cannot preserve the temporal changes.

Spatio-temporal classifier. We further analyze how different architectural choices in the classifier affect detection accuracy. We compare a plain 2D backbone (ResNet-18), ResNet-based variants with different temporal heads (LSTM [33] and Transformer [78]), and a spatio-temporal backbone based on R(2+1)D.

As illustrated in Figure 21, the best-performing variant is the R(2+1)D-based classifier, which achieves 96.7% accuracy. This result suggests that spatio-temporal modeling is particularly effective for our task, as it can capture a longer-range temporal pattern within the backbone itself. Among the ResNet-based variants, adding an LSTM head improves performance over ResNet baseline (93.3% vs. 90.0%), indicating that explicit temporal aggregation is beneficial. In contrast, the Transformer-based variant performs worse (86.7%), suggesting that a more complex temporal head does not necessarily improve performance in our setting.

Overall, all configurations achieve reasonably strong performance, suggesting that the motion signature exploited by *SweepLED* is robust across classifier designs, while spatio-temporal classifier provides the most reliable representation.

6 Discussion

We now discuss a set of key insights and considerations we find while designing and implementing *SweepLED*.

Deployment Considerations. *SweepLED*'s unobtrusive LED add-on enables several practical deployment paths. Hotels or Airbnb hosts could provide a small LED attachment in each room, much like universal chargers. Moreover, its compact form factor, low cost, and compatibility with MagSafe-style accessories make it feasible for users to carry as a lightweight travel accessory. This is practical because travelers already carry small everyday items (e.g., power banks, selfie sticks, phone cases, portable luggage scale), so *SweepLED* aligns with existing habits without adding a significant burden. Its compact and modular design also makes it well-suited for integration into accessory- or device-level hardware. For example, the LED array could be embedded into smartphone cases (e.g., Casetify, Spigen) [3, 10, 34, 69] or built into future phones that already feature notification lighting (e.g., Nothing Phone, ASUS ROG Phone) [7, 51]. These possibilities suggest that *SweepLED* can transition from a research prototype to a broadly deployable feature in commercial devices.

Concealed Cameras in Objects. From our crawl² of media reports and public disclosures of hidden camera incidents [1, 25, 74], we find that most cases involve cameras concealed inside everyday objects – such as adapters, smoke detectors, or clocks – rather than embedded directly into walls or ceilings. Motivated by this observation, we focus on object-based concealment. While *SweepLED* is not limited to this setting, emphasizing the highest-incidence scenario maximizes practical relevance.

Orthogonal and Complementary to Other Solutions. *SweepLED* is a first step toward tackling hidden camera detection from a commodity smartphone. Its optical approach is either orthogonal or complementary to other modalities. For example, LiDAR-based systems such as LAPD [62] also measure retro-reflections but face two practical barriers: (i) their camera–light coupling limits accuracy in real environments, and (ii) ToF hardware is available only on a small fraction of smartphones, limiting scalability. Likewise, RF- and EM-based methods [14, 44] detect only active cameras and miss powered-off or analog devices. A comprehensive defense will likely

²Our crawled reports of hidden camera crimes: <https://www.cyphy.kaist.ac.kr/research/sweepLED/crawled-list>

combine multiple signals, with *SweepLED* providing the optical component.

Model Training and Crowdsourcing. A key question for deployment is who maintains and updates the classification models. Several stakeholders stand to benefit – hotels, Airbnb as a platform, and authorities – and could contribute labeled data from routine inspections to enhance model generalization. With appropriate consent and safeguards, user-collected reflection signatures could also support lightweight crowdsourced [24] or federated learning [48]. Such continual updates would help *SweepLED* adapt to new lens types, concealment strategies, and environments over time.

Practical Utility and Future Directions. Recall that *SweepLED* enables fast per-object scans, requiring only 4.5 s for reliable detection (§5.2.1). For scanning-time analysis, we conservatively assume a hotel-room scenario with approximately 20 potentially suspicious objects, covering common camera-hiding categories such as smoke detectors, clocks, lamps, or chargers. This yields an upper-bound scanning time of about 1.5 minutes (4.5 s $\times$ 20 objects). Moreover, when multiple objects can be inspected within a single scan, the total scanning time can be further reduced, as demonstrated in our multi-object scanning evaluation (§5.2.2). These findings indicate that *SweepLED* already supports lightweight inspection of unfamiliar indoor environments within a short interaction window. Looking ahead, increasing the LED sweeping frequency could further suppress hand-motion jitter and enable continuous room-scale scanning, reducing user effort even further.

Limitations. *SweepLED* provides reliable detection results across diverse setups. However, its performance may degrade when a hidden camera is concealed behind heavily tinted or strongly diffusive surfaces, such as visible-range filtering films or opaque translucent layers (see A.1). For heavily tinted surfaces, this limitation may be mitigated by using wavelengths with stronger penetrative capability, such as near-infrared (NIR). For strongly diffusive materials, the challenge is more inherent: if the surface substantially diffuses the scene observed by the hidden camera, the recorded video is also likely to be severely degraded, limiting both recording quality and the temporal cues available to our method.

7 Related Work

We now present related work on hidden camera detection. Existing hidden camera detection techniques can be broadly categorized into emission-based (passive) or optical-based (active) detection. Emission-based methods operate by passively capturing emissions from devices, such as wireless signals [14, 28, 38, 43, 61, 85], electromagnetic (EM) emissions [44, 84], or heat signatures [83]. However, these methods have inherent limitations: they can only detect cameras that are currently powered on and transmitting. Consequently, devices that are inactive or configured solely for local storage evade detection. Wireless-based detection is particularly vulnerable, as cameras can easily bypass it by switching to local recording modes. Similarly, EM-based techniques suffer from degraded reliability in environments with significant EM interference, while heat-based methods can be confounded by other thermal sources in the environment.

Optical-based detection methods, on the other hand, actively illuminate an area using light sources (e.g., infrared or visible light)

and detect retro-reflection generated by camera optics. This process is typically performed either computationally on captured images [62] or human inspection of reflected light patterns (e.g., via hand-held detectors) [70]. Truong et al. [77] is an important early optical approach that demonstrates the feasibility of detecting cameras via retro-reflection and interfering with capture at the point of recording using active illumination. However, its detection stage is based on thresholding bright reflective regions in infrared images, which can also be triggered by non-camera reflective surfaces such as metallic or mirrored objects. More recently, ToF-based methods such as LAPD [62] and NIRF [82] extend optical hidden-camera detection to object-based and behind-screen scenarios, respectively. In general object-based settings, Camera–Light Coupled Multi-Angle Illumination can make observed reflection changes difficult to attribute solely to lens-specific behavior, introducing ambiguity for ordinary reflective objects. NIRF mitigates this issue in the specialized behind-screen case by exploiting a structured screen surface and ToF phase cues, but such assumptions do not readily generalize to everyday object-based hidden-camera detection. Hence, in our work, we build upon the strengths of optical-based detection while addressing these limitations in practical object-based hidden-camera detection.

8 Conclusion

We presented *SweepLED*, a practical hidden camera detection system that augments a commodity smartphone with an unobtrusive LED-embedded case. *SweepLED* leverages LED sweeping – a controlled sequence of multi-angle illumination executed while the phone remains stationary – to expose stable, lens-specific reflection cues that differentiate hidden camera lenses from ordinary reflective objects. By pairing this optical mechanism with a vision and learning pipeline, *SweepLED* achieves reliable and accurate hidden-camera detection with low user effort. We implement *SweepLED* and evaluate it in real-world environments containing concealed cameras embedded in everyday objects, along with commonly reflective non-camera items. Our experiments show that *SweepLED* achieves approximately 94% detection accuracy, a sweep time of under 5 s, and a core component cost of less than USD \$7. This demonstrates its potential as a practical and accessible tool for enhancing guest privacy in hotels and Airbnb-style short-term rentals. We provide a short demonstration of *SweepLED* in practice at: <https://www.cyphy.kaist.ac.kr/research/sweepLED>.

Acknowledgments

This research is partially supported by the National Research Foundation of Korea (NRF), funded by the Ministry of Science and ICT (MSIT) under grants RS-2024-00464269 and RS-2023-00277848. This research is also partially funded by the National Research Foundation (NRF), Singapore, and the Smart Nation & Digital Government Office (SNDGO) under its Smart Nation & Digital Government Translational R&D Grant (TRANS2023-TGC04). Any opinions, findings, conclusions, or recommendations expressed in this material are those of the author(s) and do not reflect the views of NRF and SNDGO. Jun Han is the corresponding author of this work.

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A Extended Evaluation in Diverse and Realistic Conditions

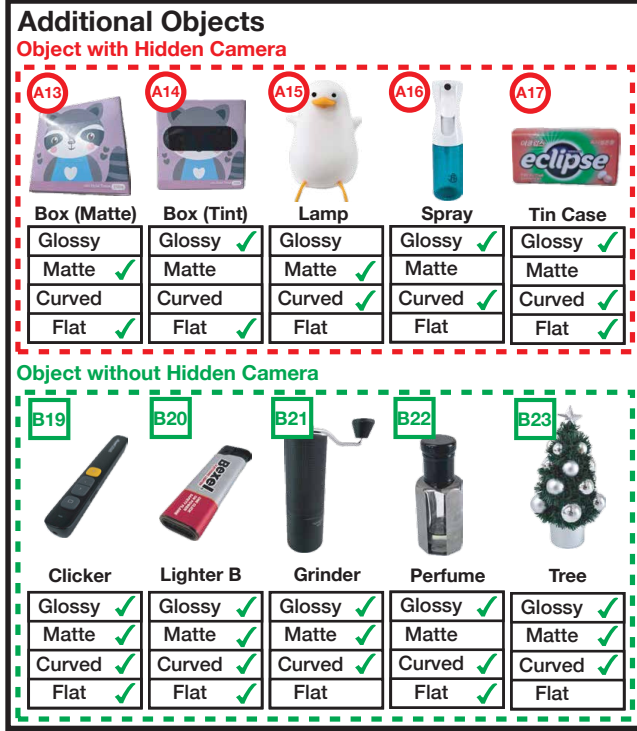


Figure 22: Figure depicts the evaluation setup for 10 additional objects. The red dotted box indicates five objects with concealed cameras, while the green dotted box indicates five additional challenging cases for detection without hidden cameras. We assign labels based on the hidden-camera region and nearby optical decoys; for objects without hidden cameras, we consider the dominant surface characteristics and the most camera-like decoy region.

We conduct additional evaluations to provide a more comprehensive analysis and further demonstrate the capabilities of *SweepLED*.

A.1 Diverse Objects Evaluation

Setup. To further evaluate robustness under diverse real-world conditions, we conduct additional experiments using 10 unseen objects, including five with hidden cameras and five without. Figure 22 illustrates these objects, which span diverse *surface reflectivity* and *geometry*.

For the objects with hidden cameras, we place the camera module behind a visually inconspicuous small hole while varying the surrounding surface geometry and reflectivity. Specifically, we prepare a tissue box with a concealed camera, and for A13 and A14, we cover the hole with either a matte surface (anti-fingerprint film) or a tinted translucent black plastic surface. For A15, A16, and A17, we create a hole within each object and position it so that it remains visually similar to the original object appearance.

For the objects without hidden cameras, we include challenging optical decoys with glossy curved surfaces, recessed metallic or laser-emitter regions (e.g., a laser-pointer clicker), and strong localized specular highlights, as these can generate compact, lens-like reflections that may confuse optical hidden-camera detectors.

These cases represent challenging scenarios in which reflections from nearby materials and object geometry may interfere with the sensing response.

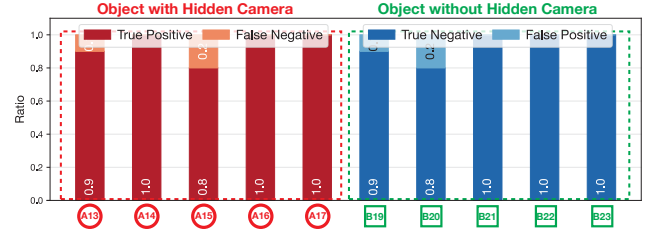


Figure 23: Figure depicts the detection results on 10 additional objects.

Results. Figure 22 shows the evaluation results. Overall, our method achieves 94.0% detection accuracy, which is broadly consistent with the results observed in our overall experiments (§5.2). While performance shows a slight decrease in a few cases, the system still demonstrates consistently reliable performance across all tested objects. We also observe that objects such as B19 and B20, which contain recessed structures with laser or metallic components, can produce reflection patterns similar to those of hidden cameras. Nevertheless, our method remains robust under these conditions and maintains accurate detection. These results suggest that the proposed system generalizes well to diverse unseen objects, although incorporating even harsher conditions into the dataset would provide a more rigorous evaluation of robustness.

We also identify several particularly challenging cases. First, *SweepLED*'s performance degrades under highly adverse conditions in which the covering material is strongly diffusive. In such cases, the covering severely scatters the retro-reflections from the camera lens, hindering detection. At the same time, it also degrades the image captured by the hidden camera itself, leaving only limited visual information. As a result, these conditions impair imaging not only for our method but also for the hidden camera attempting to observe the scene. Even when *SweepLED* fails to detect the camera, the captured image is often significantly blurred, so that the object shape is barely recognizable, suggesting that the hidden camera would likewise recover little useful information about the scene.

Second, detection becomes nearly impossible when the concealed region is covered by a tinted translucent film that does not allow sufficient light transmission in the visible range. This limitation arises not merely from the sensing algorithm, but from the physical attenuation and blockage of light itself. We envision that extending our mechanism to more penetrative wavelengths could enable detection in such scenarios, potentially overcoming the physical limitations imposed by optically obstructive materials in the visible spectrum.



Figure 24: Figure depicts experimental setups for evaluating the impact of different user postures, including standing (2.0m), and overhead (2.4m) scenarios. These setups differ in target distance by approximately 0.8 m.

A.2 Different Postures

We conduct an additional evaluation under different usage postures to examine the practicality of *SweepLED* in diverse settings. Since hidden cameras may be installed at various locations, such as smoke detectors mounted on ceilings or motion sensors attached to walls, we evaluate *SweepLED* across multiple deployment setups.

As shown in Figure 24, we place a motion sensor on a wall at a height of 2.0m and a gas detector on the ceiling at a height of 2.4m to approximate a typical lodging scenario. We then compare the results from these two postures with those from the sitting posture used in our main usage scenario. The results show that *SweepLED* maintains the same level of performance across different postures. Specifically, detection accuracy remains 100% for both the wall-mounted motion sensor and the ceiling-mounted gas detector, matching the performance observed in the sitting posture.